NEUTRALIZATION PROBLEMS

1. What volume of a 0.200 M NaOH solution is necessary to:

   a) neutralize 25.0 mL of a 0.500 M HCl solution? (62.5 mL)

   b) neutralize 23.1 mL of a 0.100 M H_2SO_4 solution? (23.1 mL)

2. It is found that 26.5 mL of a 0.050 M sulphuric acid solution (H_2SO_4) will neutralize 10.0 mL of a NaOH solution. What is the molarity of the sodium hydroxide solution?

   (0.27 M)

3. A sulphuric acid solution is used to neutralize a barium hydroxide solution. It is found that exactly 56.7 mL of 1.2 M sulphuric acid can neutralize 34.2 mL of barium hydroxide, $Ba(OH)_2$. What is the concentration of the barium hydroxide?

   (1.99 M)

4. 25.0 mL of a 0.50 M Ca(OH)2 solution neutralized 10.0 mL of a phosphoric acid solution, H_3PO_4. What is the equation for the neutralization reaction? What is the molarity of phosphoric acid?

   (0.83 M)